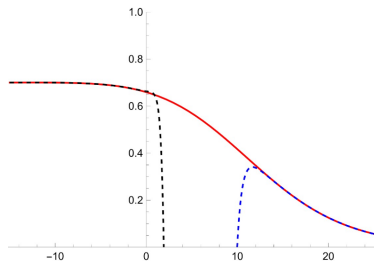
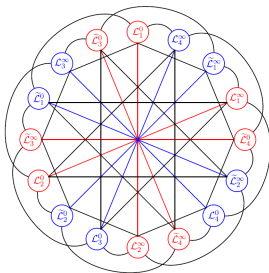


On monodromy surfaces and asymptotics of q -Painlevé equations

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Seminar at Waseda University, Sep 2026

Collaborators



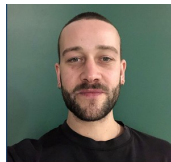
Harini Desiraju



Nalini Joshi



Marta Mazzocco



Alexander Stokes

Based on papers:

- (1) *On the monodromy manifold of q -Painlevé VI*
with Nalini Joshi, CMP, '23.
- (2) *On q -Painlevé VI and the geometry of Segre surfaces*
Nonlinearity, '24.
- (3) *Segre surfaces and the geometry of Painlevé equations*,
with Nalini Joshi and Marta Mazzocco, CMP, '26.
- (4) *A tau function for q -Painlevé VI as a Fredholm determinant*,
with Harini Desiraju, arXiv:2608.03345, '26.
- (5) *On monodromy of monodromy surfaces*,
with Alexander Stokes, arXiv:2608.25594, '26.

Plan talk

1 Part I: the classical story

- Painlevé equations
- Monodromy surfaces
- Asymptotic analysis

2 Part II: the q -analog story

- q -Painlevé equations
- Monodromy surface of qP_{VI}
- Asymptotic analysis of qP_{VI}

3 Part III: monodromy surfaces revisited

- Monodromy surfaces as embedded affine del Pezzo surfaces
- Counterpart of Sakai's classification under Riemann-Hilbert?

Special functions and linear ODEs

- Many of the classical special functions (Whittaker/Bessel functions, Hermite polynomials etc), satisfy a **linear** second order ODE,

$$y''(z) + a(z)y'(z) + b(z)y(z) = 0.$$

- Linearity** $\implies$ **singularities are fixed**: Singularities of solutions are a subset of singularities of coefficients of ODE.
- For example, solutions of the **hypergeometric equation**,

$$y''(z) + \frac{z(\alpha + \beta + 1) - \gamma}{z(z-1)}y'(z) + \frac{\alpha\beta}{z(z-1)}y(z) = 0,$$

only have singularities at $z = 0, 1, \infty$.

- On the other hand, considering **nonlinear ODEs**,

$$u_{tt} = R(u, u_t, t), \quad R \text{ rational},$$

solutions generally have **movable branch points**, i.e. their locations vary per solution and cannot be read off the ODE itself.

Painlevé equations: hunt for nonlinear special functions

Painlevé, Gambier and Picard (~ 1900), set out to classify all **nonlinear second order ODEs** that share the nice property with linear ODEs that the locations of singularities of solutions are fixed.

Result of classification:

$$P_I : \quad u_{tt} = 6u^2 + t,$$

$$P_{II} : \quad u_{tt} = 2u^3 + tu + \alpha,$$

$$P_{III} : \quad u_{tt} = \frac{1}{u}u_t^2 - \frac{1}{t}u_t + \frac{1}{t}(\alpha u^2 + \beta) + \gamma u^3 + \frac{\delta}{u},$$

$$P_{IV} : \quad u_{tt} = \frac{1}{2u}u_t^2 + \frac{3}{2}u^3 + 4tu^2 + 2(t^2 - \alpha)u + \frac{\beta}{u},$$

$$P_V : \quad u_{tt} = \left(\frac{1}{2u} + \frac{1}{u-1} \right) u_t^2 - \frac{1}{t}u_t + \frac{(u-1)^2}{t^2} \left(\alpha u + \frac{\beta}{u} \right) + \frac{\gamma}{t}u + \delta \frac{u(u+1)}{u-1},$$

$$P_{VI} : \quad u_{tt} = ?$$

Here $\alpha, \beta, \gamma, \delta \in \mathbb{C}$ are parameters.

The sixth Painlevé equation

- Painlevé, Gambier and Picard missed P_{VI} in their classification.
- R. Fuchs discovered P_{VI} (1905),

$$u_{tt} = \left(\frac{1}{u} + \frac{1}{u-1} + \frac{1}{u-t} \right) \frac{u_t^2}{2} - \left(\frac{1}{t} + \frac{1}{t-1} + \frac{1}{u-t} \right) u_t \\ + \frac{u(u-1)(u-t)}{2t^2(t-1)^2} \left((2\theta_\infty - 1)^2 - \frac{4\theta_0^2 t}{u^2} + \frac{4\theta_1^2(t-1)}{(u-1)^2} + \frac{(1-4\theta_t^2)t(t-1)}{(u-t)^2} \right).$$

- P_{VI} sits at the top of the Painlevé hierarchy, with four complex parameters $\theta := (\theta_0, \theta_t, \theta_1, \theta_\infty)$, from which the others can be obtained through degeneration:

$$\begin{array}{ccccc} P_{VI} & \longrightarrow & P_V & \longrightarrow & P_{III} \\ & & \downarrow & & \downarrow \\ & & P_{IV} & \longrightarrow & P_{II} \longrightarrow P_I \end{array}$$

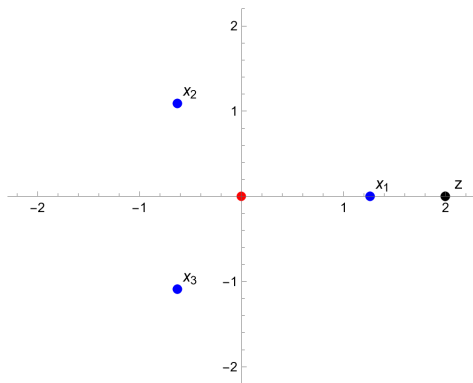
- Fuchs discovered P_{VI} whilst studying **deformation** of a second order **linear** ODE that **preserves monodromy** (integrability).

Origins of Monodromy

Monodromy originates in Riemann's work (~1850s) on algebraic functions, complex integrals and ODEs over $\mathbb{C}$.

It describes how objects change when analytically continued along closed loops that encircle **singularities**.

Solutions of $x^3 - z = 0$ are cyclically permuted as z goes around $z = 0$.



$$x_1 = \sqrt[3]{z}$$

$$x_2 = \zeta_3 \sqrt[3]{z}$$

$$x_3 = \zeta_3^{-1} \sqrt[3]{z}$$

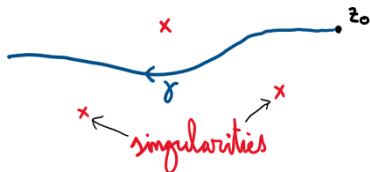
Solving ODEs along curves

Consider a linear ODE

$$\frac{dY}{dz} = A(z)Y, \quad (z \in \mathbb{C}),$$

where $A(z)$ a matrix of rational functions over $\mathbb{C}$.

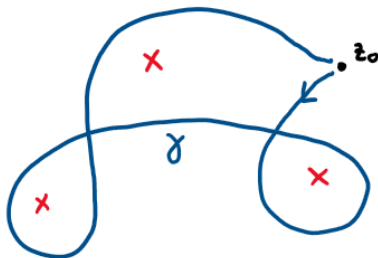
- Both A and Y are $n \times n$ matrices here.
- Given a curve γ starting at a point z_0 that avoids **singularities** of $A(z)$,



and an initial value $Y(z_0) = Y_0 \in GL_n(\mathbb{C})$, the ODE can be uniquely solved along γ .

Monodromy of an ODE

If we pick a closed loop γ ,



we get the **monodromy matrix**

$$M_\gamma = Y(\text{start})^{-1} Y(\text{end}).$$

This defines the **monodromy representation**

$$\pi_1(\mathbb{CP}^1 - \{\text{singularities}\}, z_0) \rightarrow GL_n(\mathbb{C}), \gamma \mapsto M_\gamma.$$

Monodromy preserving deformation

Fuchs' work shows that Painlevé VI governs **deformation** in the class of rank 2 Fuchsian ODEs with four singularities ($z = 0, t, 1, \infty$),

$$\frac{dY}{dz} = \left(\frac{A_0}{z} + \frac{A_t}{z-t} + \frac{A_1}{z-1} \right) Y, \quad A_0, A_t, A_1 \in \mathfrak{sl}_2(\mathbb{C}),$$

that **preserves monodromy**.

Monodromy representations (up to overall conjugation) of this class of ODEs form a 4-parameter family of **cubic surfaces**,

$$\mathcal{M}_\theta : \mathbf{x}_1 \mathbf{x}_2 \mathbf{x}_3 + \mathbf{x}_1^2 + \mathbf{x}_2^2 + \mathbf{x}_3^2 - w_1 \mathbf{x}_1 - w_2 \mathbf{x}_2 - w_3 \mathbf{x}_3 + w_4 = 0,$$

where, writing $y_j = 2 \cos(2\pi\theta_j)$ for $j \in \{0, t, 1, \infty\}$,

$$\begin{aligned} w_1 &= y_0 y_\infty + y_t y_1, & w_2 &= y_t y_\infty + y_0 y_1, \\ w_3 &= y_1 y_\infty + y_0 y_t, & w_4 &= y_0^2 + y_t^2 + y_1^2 + y_\infty^2 + y_0 y_t y_1 y_\infty - 4. \end{aligned}$$

The quantities $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ are **integrals of motion** of P_{VI} , lying on the **monodromy surface** $\mathcal{M}_\theta$, also called the **Jimbo-Fricke cubic**.

Monodromy surfaces

For each Painlevé equation P_K , a corresponding monodromy surface $\mathcal{M}_K$ can be realised as an **affine cubic surface**

$$\mathcal{M}_K \cong \{x \in \mathbb{C}^3 : R_K(x) = 0\} \quad (R_K \text{ a cubic polynomial}).$$

P-eqs	polynomials
P_{VI}	$x_1 x_2 x_3 + x_1^2 + x_2^2 + x_3^2 - w_1 x_1 - w_2 x_2 - w_3 x_3 + w_4$
P_V	$x_1 x_2 x_3 + x_1^2 + x_2^2 - w_1 x_1 - w_2 x_2 - x_3 + w_3$
P_V^{deg}	$x_1 x_2 x_3 + x_1^2 + x_2^2 + w_1 x_1 + w_2 x_2 + 1$
P_{IV}	$x_1 x_2 x_3 + x_1^2 - w_1 x_1 - x_2 - x_3 + w_2$
$P_{III}^{D_6}$	$x_1 x_2 x_3 + x_1^2 + x_2^2 - w_1 x_1 - w_2 x_2 + 1$
$P_{III}^{D_7}$	$x_1 x_2 x_3 + x_1^2 + x_2^2 - w_1 x_1 - x_2$
$P_{III}^{D_8}$	$x_1 x_2 x_3 + x_1^2 + x_2^2 - x_1$
P_{II}^{JM}	$x_1 x_2 x_3 + x_1^2 - w_1 x_1 - x_2 - 1$
P_{II}^{FN}	$x_1 x_2 x_3 - x_1 - x_2 - x_3 + w_1$
P_I	$x_1 x_2 x_3 - x_1 - x_2 + 1$

See e.g. van der Put and Saito (2009) and Chekhov et al. (2015).

Riemann-Hilbert correspondence

- Riemann (1857) had the fundamental insight that monodromy representations determine regular ODEs uniquely up to equivalence.
- Hilbert's 21st problem (1900) asks whether any monodromy representation can be realised by a Fuchsian ODE.
- Algebraic treatment pioneered by Deligne (1970) lead to **Riemann-Hilbert correspondence**. In the simplest setting:

$$\left\{ \begin{array}{l} \text{Regular ODEs on } \mathbb{CP}^1 \\ \text{with given singularities} \end{array} \right\} \leftrightarrow \left\{ \begin{array}{l} \text{points on a} \\ \text{character variety} \end{array} \right\}.$$

- Malgrange (80's) extended correspondence on $\mathbb{CP}^1$ to **irregular ODEs**, enriching monodromy data with Stokes data. Corollary:

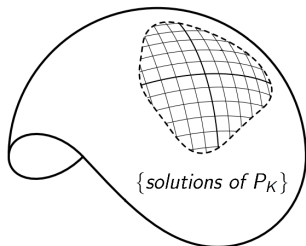
$$\text{RH: } \left\{ \begin{array}{l} \text{solutions of a} \\ \text{Painlevé equation} \end{array} \right\} \leftrightarrow \left\{ \begin{array}{l} \text{points on a} \\ \text{monodromy surface} \end{array} \right\}.$$

Global coordinates

Writing

$$u(t) = u(t; \theta_0, \theta_t, \theta_1, \theta_\infty, \mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3),$$

One can think of $(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3) \in \mathcal{M}_\theta$ as **coordinates** of the solution space of $P_{VI}(\theta)$.



Computing the value of $u(t)$ at any given $t \in \mathbb{C} \setminus \{0, 1\}$ is only possible numerically, but we can describe $u(t)$ **asymptotically** as $t \rightarrow 0$, $t \rightarrow 1$ or $t \rightarrow \infty$.

Jimbo's asymptotic formulas (1982)

For generic $\mathbf{x} \in \mathcal{M}_\theta$, Jimbo's (1982) asymptotic formulas for P_{VI} tau-function imply

$$u(t) \sim \frac{u_{0t}}{c_{0t} s_{0t}} (-t)^{1-2\sigma_{0t}} \quad (t \rightarrow 0, |\arg(-t)| < \pi),$$

where $2 \cos(2\pi\sigma_{0t}) = \mathbf{x}_3$,

$$u_{0t} = + \frac{(\theta_0 + \theta_t + \sigma_{0t})(-\theta_0 + \theta_t + \sigma_{0t})(\theta_\infty + \theta_1 + \sigma_{0t})}{4(\theta_\infty + \theta_1 - \sigma_{0t})\sigma_{0t}^2},$$

$$c_{0t} = \frac{\Gamma(1-2\sigma_{0t})^2}{\Gamma(1+2\sigma_{0t})^2} \prod_{\epsilon=\pm 1} \frac{\Gamma(1+\theta_t+\epsilon\theta_0+\sigma_{0t})\Gamma(1+\theta_1+\epsilon\theta_\infty+\sigma_{0t})}{\Gamma(1+\theta_t+\epsilon\theta_0-\sigma_{0t})\Gamma(1+\theta_1+\epsilon\theta_\infty-\sigma_{0t})},$$

and

$$s_{0t} = \frac{a + b e^{2\pi i \sigma_{0t}}}{d},$$

$$a = \frac{1}{2} i \sin(2\pi\sigma_{0t}) \mathbf{x}_1 + \cos(2\pi\theta_0) \cos(2\pi\theta_1) + \cos(2\pi\theta_\infty) \cos(2\pi\theta_t),$$

$$b = \frac{1}{2} i \sin(2\pi\sigma_{0t}) \mathbf{x}_2 + \cos(2\pi\theta_t) \cos(2\pi\theta_1) + \cos(2\pi\theta_\infty) \cos(2\pi\theta_0),$$

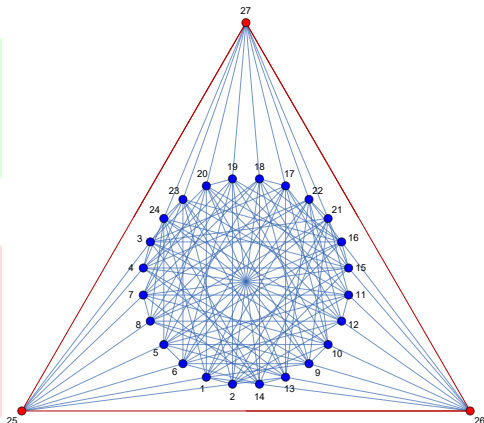
$$d = 4 \prod_{\epsilon=\pm 1} \sin[\pi(\theta_0 + \epsilon(\theta_t - \sigma_{0t}))] \sin[\pi(\theta_\infty + \epsilon(\theta_1 - \sigma_{0t}))],$$

24 lines on $\mathcal{M}_\theta$

Cayley and Salmon (1849) showed that every smooth cubic surface contains exactly **27 lines**.

$\overline{\mathcal{M}_\theta} \setminus \mathcal{M}_\theta$ consists of 3 lines.

On each of the 24 lines in $\mathcal{M}_\theta$, one of the general asymptotic expansions around $t = 0, 1$ and ∞ **truncates** (Guzzetti '06).



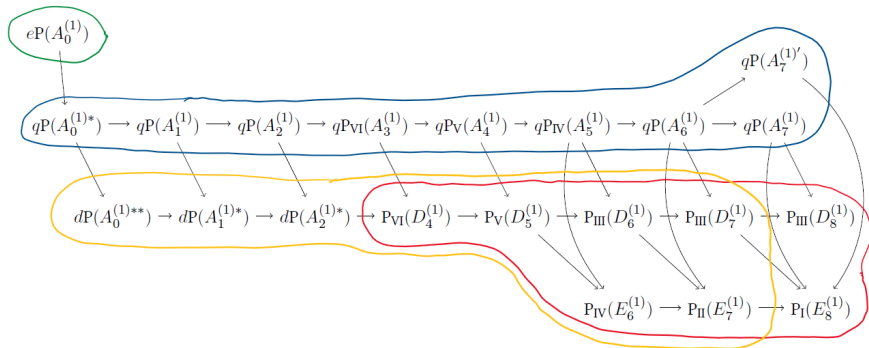
Conjecture (Ramis, '22)

Every **line** of every monodromy surface (for all the differential Painlevé equations and for all the values of the parameters, generic or not) is a one parameter family of “**truncated**” solutions.

Conjecture has been verified for P_{VI} , P_{IV} , P_{II} , P_I .

Discrete Painlevé equations

Sakai (2001) classified all Painlevé equations, differential and discrete, in terms of their initial value spaces.



- In **green** the elliptic Painlevé equation.
- In **blue** the q -difference Painlevé equations.
- In **yellow** the additive Painlevé equations.
- In **red** the classical Painlevé equations.

q -Painlevé VI

Let $q \in \mathbb{C}$ with $0 < |q| < 1$, then q -Painlevé VI is given by

$$qP_{VI}(\theta, t_0) : \begin{cases} f\bar{f} = \frac{(\bar{g} - q^{+\theta_0}t)(\bar{g} - q^{-\theta_0}t)}{(\bar{g} - q^{\theta_\infty-1})(\bar{g} - q^{-\theta_\infty})}, \\ g\bar{g} = \frac{(f - q^{+\theta_t}t)(f - q^{-\theta_t}t)}{q(f - q^{+\theta_1})(f - q^{-\theta_1})}, \end{cases}$$

where

- $f, g : T \rightarrow \mathbb{CP}^1$ are complex functions on a discrete time domain

$$T = q^{\mathbb{Z}}t_0 := \{\dots, q^{+2}t_0, q^{+1}t_0, t_0, q^{-1}t_0, q^{-2}t_0, \dots\},$$

and t varies in this domain.

- $f = f(t)$, $\bar{f} = f(qt)$, and similar for g ,
- $\theta = (\theta_0, \theta_t, \theta_1, \theta_\infty) \in \mathbb{C}^4$ and $t_0 \in \mathbb{C}^*$ are complex parameters.

Discovery of qP_{VI}

Jimbo and Sakai (1996) derived qP_{VI} as governing **deformation** of a rank and degree two Fuchsian q -difference system which leaves **monodromy** invariant.

After normalisation, this linear system takes the form

$$Y(qz) = A(z, t)Y(z), \quad A(z, t) = A_0 + z A_1 + z^2 A_2,$$

where

$$A_0 = H q^{\theta_0 \sigma_3} t H^{-1}, \quad A_2 = q^{-\theta_\infty \sigma_3}, \quad \sigma_3 := \text{diag}(1, -1),$$

for some $H = H(t) \in GL_2(\mathbb{C})$, and

$$|A(z, t)| = (z - q^{+\theta_t} t)(z - q^{-\theta_t} t)(z - q^{+\theta_1})(z - q^{-\theta_1}).$$

Monodromy

Following Birkhoff (1913) and Sauloy (2002), the **monodromy** of such a system is encapsulated by a single **connection matrix**,

$$C(z) := \Psi_0(z)^{-1} \Psi_\infty(z),$$

where $\Psi_0(z)$ and $\Psi_\infty(z)$ are meromorphic matrix functions on $\mathbb{CP}^1 \setminus \{\infty\}$ and $\mathbb{CP}^1 \setminus \{0\}$ resp. that define canonical solutions around $z = 0$ and $z = \infty$,

$$\begin{aligned} Y_0(z) &= z^{\log_q(t)} \Psi_0(z) z^{\theta_0 \sigma_3}, & \Psi_0(z) &= H + \mathcal{O}(z) \quad (z \rightarrow 0), \\ Y_\infty(z) &= z^{\log_q(z/q)} \Psi_\infty(z) z^{-\theta_\infty \sigma_3}, & \Psi_\infty(z) &= I + \mathcal{O}(z^{-1}) \quad (z \rightarrow \infty), \end{aligned}$$

of the linear system $Y(qz) = A(z)Y(z)$.

Space of monodromy data

The connection matrix $C(z)$ satisfies

- (1) $C(z)$ is analytic on $\mathbb{C}^*$.
- (2) $C(qz) = t z^{-2} q^{\theta_0 \sigma_3} C(z) q^{\theta_\infty \sigma_3}$.
- (3) $|C(z)| = \text{constant} \times \theta_q(q^{-\theta_t \frac{z}{t}}) \theta_q(q^{+\theta_t \frac{z}{t}}) \theta_q(q^{-\theta_1 z}) \theta_q(q^{+\theta_1 z})$,

where $\theta_q(\cdot)$ denotes the modified Jacobi theta function.

Define the **space of monodromy data** $M_q(\Theta, t)$ as the space of matrices $C(z)$ satisfying (1)-(3), quotiented by arbitrary left and right-multiplication by constant invertible diagonal matrices.

This space was first introduced and studied by Ohyama, Ramis and Sauloy (2020). They showed that it naturally comes with the structure of an algebraic variety.

Tyurin parameters

For any 2×2 matrix R of rank 1, define $\pi(R) \in \mathbb{CP}^1$ by

$$R_1 = \pi(R)R_2, \quad R = (R_1, R_2).$$

To construct integrals of motion, we use **Tyurin parameters** of the connection matrix,

$$\rho_k = \pi(C(u_k)) \quad (1 \leq k \leq 4), \quad (u_1, u_2, u_3, u_4) = (q^{+\theta_t}t, q^{-\theta_t}t, q^{+\theta_1}, q^{-\theta_1}),$$

The Tyurin parameters satisfy

$$T_{12}\rho_1\rho_2 + T_{13}\rho_1\rho_3 + T_{14}\rho_1\rho_4 + T_{23}\rho_2\rho_3 + T_{24}\rho_2\rho_4 + T_{34}\rho_3\rho_4 = 0,$$

$$T'_{12}\rho_1\rho_2 + T'_{13}\rho_1\rho_3 + T'_{14}\rho_1\rho_4 + T'_{23}\rho_2\rho_3 + T'_{24}\rho_2\rho_4 + T'_{34}\rho_3\rho_4 \neq 0,$$

where, for any labeling $\{i, j, k, l\} = \{1, 2, 3, 4\}$,

$$T_{ij} = \frac{u_2 u_4}{q^{\theta_0 + \theta_\infty} t} u_i u_l \theta_q \left(\frac{u_i}{u_j}, \frac{u_k}{u_l}, \frac{u_i u_j}{q^{+\theta_0 - \theta_\infty} t}, \frac{u_k u_l}{q^{+\theta_0 + \theta_\infty} t} \right), \quad T'_{ij} = T_{ij}|_{\theta_0=0}.$$

Integrals of motion

For any $1 \leq i < j \leq 4$,

$$\mathbf{x}_{ij} = \frac{T_{ij}\rho_i\rho_j}{T'_{12}\rho_1\rho_2 + T'_{13}\rho_1\rho_3 + T'_{14}\rho_1\rho_4 + T'_{23}\rho_2\rho_3 + T'_{24}\rho_2\rho_4 + T'_{34}\rho_3\rho_4},$$

defines an **integral of motion** of qP_{VI} .

Theorem (Joshi and PR ('23))

The six integrals of motion,

$$\mathbf{x} = (\mathbf{x}_{12}, \mathbf{x}_{13}, \mathbf{x}_{14}, \mathbf{x}_{23}, \mathbf{x}_{24}, \mathbf{x}_{34}),$$

lie on an explicit affine algebraic surface $\mathcal{M}_{q,\theta,t_0}$, see next slide.

The induced mapping

$$\mathrm{RH}_q : \{\text{Solutions of } qP_{VI}(\theta, t_0)\} \rightarrow \mathcal{M}_{q,\theta,t_0}, (f, g) \mapsto \mathbf{x},$$

is a one-to-one correspondence, for generic values of the parameters.

In fact, RH_q is a **biholomorphism** when identifying the solution space of $qP_{VI}(\theta, t_0)$ with the initial value space at $t = t_0$ [PR 24].

Monodromy surface $\mathcal{M}_{q,\theta,t_0}$

The algebraic surface $\mathcal{M}_{q,\theta,t_0}$ is defined in $\{x \in \mathbb{C}^6\}$, by the equations

$$x_{12} + x_{13} + x_{14} + x_{23} + x_{24} + x_{34} = 0,$$

$$a_{12}x_{12} + a_{13}x_{13} + a_{14}x_{14} + a_{23}x_{23} + a_{24}x_{24} + a_{34}x_{34} + a_{\infty} = 0,$$

$$x_{13}x_{24} - b_1x_{12}x_{34} = 0,$$

$$x_{14}x_{23} - b_2x_{12}x_{34} = 0,$$

where

$$a_{12} = \prod_{\epsilon=\pm 1} \frac{\theta_q(q^{+\theta_{\infty}}t_0)}{\theta_q(q^{\epsilon\theta_0+\theta_{\infty}}t_0)},$$

$$a_{34} = \prod_{\epsilon=\pm 1} \frac{\theta_q(q^{-\theta_{\infty}}t_0)}{\theta_q(q^{\epsilon\theta_0-\theta_{\infty}}t_0)},$$

$$a_{13} = \prod_{\epsilon=\pm 1} \frac{\vartheta_q(\theta_t + \theta_1 + \theta_{\infty})}{\vartheta_q(\epsilon\theta_0 + \theta_t + \theta_1 + \theta_{\infty})},$$

$$a_{24} = \prod_{\epsilon=\pm 1} \frac{\vartheta_q(-\theta_t - \theta_1 + \theta_{\infty})}{\vartheta_q(\epsilon\theta_0 - \theta_t - \theta_1 + \theta_{\infty})},$$

$$a_{14} = \prod_{\epsilon=\pm 1} \frac{\vartheta_q(\theta_t - \theta_1 + \theta_{\infty})}{\vartheta_q(\epsilon\theta_0 + \theta_t - \theta_1 + \theta_{\infty})},$$

$$a_{23} = \prod_{\epsilon=\pm 1} \frac{\vartheta_q(-\theta_t + \theta_1 + \theta_{\infty})}{\vartheta_q(\epsilon\theta_0 - \theta_t + \theta_1 + \theta_{\infty})},$$

and similar expressions for a_{∞}, b_1, b_2 , where $\vartheta_q(x) = \theta_q(q^x)$.

Segre surfaces

The surface $\mathcal{M}_{q,\theta,t_0}$ is an affine **Segre surface**.

A Segre surface is by definition the intersection of two quadrics in $\mathbb{CP}^4$,

$$\{x \in \mathbb{CP}^4 : P(x) = 0\} \cap \{x \in \mathbb{CP}^4 : Q(x) = 0\},$$

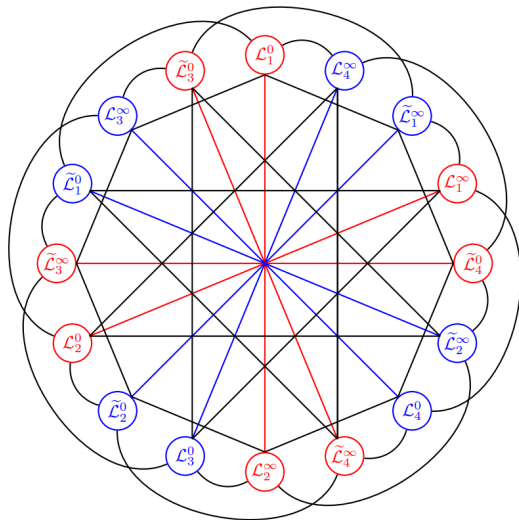
where P and Q quadratic polynomials.

They were introduced and studied by Corrado Segre (1884).

What can geometry tell us?

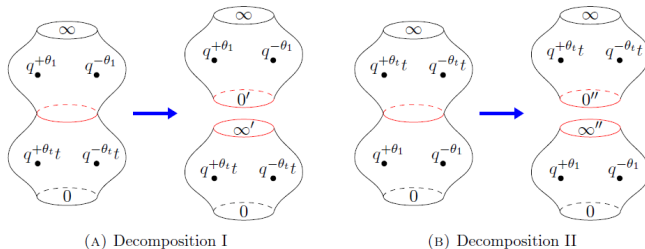
Theorem (Corrado Segre (1884))

*A smooth Segre surface contains exactly **16 lines**.*



- vertices : lines
- edges : intersection points

Mano-decompositions



Writing $t_m = q^m t_0$, a connection matrix admits **Mano-decompositions**

$$\begin{aligned} z^m C(z) &= C_l^i(z/t_m) (-t_m)^{\sigma_{0t} \sigma_3} C_l^e(z), \\ &= C_{ll}^i(z) (-t_m)^{-\sigma_{01} \sigma_3} C_{ll}^e(z/t_m), \end{aligned}$$

where the components are **Heine hypergeometric** connection matrices.

Such decompositions were first observed in Mano's asymptotic study (2010) of qP_{VI} . Proven in general by Ohyama, Ramis and Sauloy (2020).

Lines correspond to **reducible factors** in Mano-decompositions.

Generic Asymptotics

Theorem (PR '24)

Take a generic $\mathbf{x} \in \mathcal{M}_{q,\theta,t_0}$, then the corresponding solution (f, g) of $qP_{VI}(\theta, t_0)$ admits simultaneous complete asymptotic expansions,

$$f(t) = \sum_{n=1}^{\infty} \sum_{k=-n}^n F_{n,k} r_{0t}^k (-t)^{n+2k\sigma_{0t}},$$
$$g(t) = \sum_{n=1}^{\infty} \sum_{k=-n}^n G_{n,k} r_{0t}^k (-t)^{n+2k\sigma_{0t}},$$

absolutely convergent for small enough $t \in q^{\mathbb{Z}}t_0$, and

$$\frac{f(t)}{t} = \sum_{n=1}^{\infty} \sum_{k=-n}^n \dot{F}_{n,k} r_{01}^k (-t)^{-(n+2k\sigma_{01})},$$
$$\frac{1}{g(t)} = \sum_{n=1}^{\infty} \sum_{k=-n}^n \dot{G}_{n,k} r_{01}^k (-t)^{-(n+2k\sigma_{01})},$$

absolutely convergent for large enough $t \in q^{\mathbb{Z}}t_0$, with integration constants $\{\sigma_{0t}, r_{0t}\}$ and $\{\sigma_{01}, r_{01}\}$ as explicit functions of $\mathbf{x}$ and θ, t_0 .

Some explicit formulas

The exponents are defined through

$$\begin{aligned}\frac{\vartheta_q(\sigma_{0t} - \theta_1 + \theta_\infty)\vartheta_q(\sigma_{0t} + \theta_1 - \theta_\infty)}{\vartheta_q(\sigma_{0t} + \theta_1 + \theta_\infty)\vartheta_q(\sigma_{0t} - \theta_1 - \theta_\infty)} &= \frac{T_{14}\mathbf{x}_{13}}{T_{13}\mathbf{x}_{14}}, \\ \frac{\vartheta_q(\sigma_{01} - \theta_t + \theta_\infty)\vartheta_q(\sigma_{01} + \theta_t - \theta_\infty)}{\vartheta_q(\sigma_{01} + \theta_t + \theta_\infty)\vartheta_q(\sigma_{01} - \theta_t - \theta_\infty)} &= \frac{T_{23}\mathbf{x}_{13}}{T_{13}\mathbf{x}_{23}}, \\ 0 &< \Re \sigma_{0t}, \Re \sigma_{01} < \frac{1}{2},\end{aligned}$$

where $\vartheta_q(x) = \theta_q(q^x)$, and

$$\begin{aligned}r_{0t} &= c_{0t} \times s_{0t}, & s_{0t} &= M_{0t} \left(\frac{T_{34}\mathbf{x}_{23}}{T_{23}\mathbf{x}_{34}} \right), \\ r_{01} &= c_{01} \times s_{01}, & s_{01} &= M_{01} \left(\frac{T_{34}\mathbf{x}_{23}}{T_{23}\mathbf{x}_{34}} \right),\end{aligned}$$

where $M_{0t}(\cdot)$ and $M_{01}(\cdot)$ are some explicit Möbius transforms and

$$\begin{aligned}c_{0t} &= \frac{\Gamma_q(1 - 2\sigma_{0t})^2}{\Gamma_q(1 + 2\sigma_{0t})^2} \prod_{\epsilon=\pm 1} \frac{\Gamma_q(1 + \theta_t + \epsilon\theta_0 + \sigma_{0t})\Gamma_q(1 + \theta_1 + \epsilon\theta_\infty + \sigma_{0t})}{\Gamma_q(1 + \theta_t + \epsilon\theta_0 - \sigma_{0t})\Gamma_q(1 + \theta_1 + \epsilon\theta_\infty - \sigma_{0t})}, \\ c_{01} &= \frac{\Gamma_q(1 - 2\sigma_{01})^2}{\Gamma_q(1 + 2\sigma_{01})^2} \prod_{\epsilon=\pm 1} \frac{\Gamma_q(1 + \theta_1 + \epsilon\theta_0 + \sigma_{01})\Gamma_q(1 + \theta_t + \epsilon\theta_\infty + \sigma_{01})}{\Gamma_q(1 + \theta_1 + \epsilon\theta_0 - \sigma_{01})\Gamma_q(1 + \theta_t + \epsilon\theta_\infty - \sigma_{01})}.\end{aligned}$$

Leading order coefficients

Leading order coefficients in asymptotic expansion near $t = 0$,

$$F_{1,\pm 1} = q^{-\theta_t} \frac{(q^{\theta_t + \theta_0 \mp \sigma_{0t}} - 1)(q^{\theta_t - \theta_0 \mp \sigma_{0t}} - 1)(q^{\theta_1 + \theta_\infty \mp \sigma_{0t}} - 1)}{(q^{\theta_1 + \theta_\infty \pm \sigma_{0t}} - 1)(q^{\sigma_{0t}} - q^{-\sigma_{0t}})^2},$$

$$F_{1,0} = \frac{2(q^{\theta_t} + q^{-\theta_t}) - (q^{\theta_0} + q^{-\theta_0})(q^{\sigma_{0t}} + q^{-\sigma_{0t}})}{(q^{\sigma_{0t}} - q^{-\sigma_{0t}})^2},$$

$$G_{1,0} = \frac{2(q^{\theta_0} + q^{-\theta_0}) - (q^{\theta_t} + q^{-\theta_t})(q^{\sigma_{0t}} + q^{-\sigma_{0t}})}{(q^{\sigma_{0t}} - q^{-\sigma_{0t}})^2} q^{-1},$$

$$G_{1,\pm 1} = -q^{-1 \mp \sigma_{0t}} F_{1,\pm 1},$$

and similar formulas near $t = \infty$, e.g.

$$\dot{F}_{1,\pm 1} = q^{-\theta_1} \frac{(q^{\theta_1 + \theta_0 \mp \sigma_{01}} - 1)(q^{\theta_1 - \theta_0 \mp \sigma_{01}} - 1)(q^{\theta_t + \theta_\infty \mp \sigma_{01}} - 1)}{(q^{\theta_t + \theta_\infty \pm \sigma_{01}} - 1)(q^{\sigma_{01}} - q^{-\sigma_{01}})^2},$$

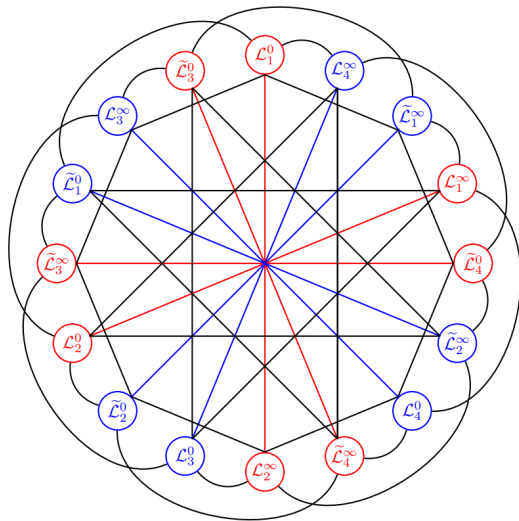
A short history of asymptotic studies

- Mano ('10): generic leading order asymptotics near $t = 0$ and $t = \infty$ of qP_{VI} solutions and implicit relations between them.
- Jimbo, Nagoya and Sakai ('17): conjectural complete (and fully explicit) asymptotic expansion near $t = 0$ of the generic qP_{VI} tau-function, from conformal field theory.
- PR ('24): complete asymptotic expansions near $t = 0$ and $t = \infty$ of qP_{VI} solutions with explicit nonlinear connection formulas.
- Desiraju and PR ('26): asymptotic expansion near $t = 0$ of the generic qP_{VI} tau-function from Riemann-Hilbert theory.

What can geometry tell us?

Theorem (Corrado Segre (1884))

*A generic Segre surface contains exactly **16 lines**.*



- vertices : lines
- edges : intersection points

Truncation on lines

On the **blue lines** the generic asymptotics near $t = 0$ **truncate**.

For example, on the line $\tilde{\mathcal{L}}_2^\infty$, we have $\sigma_{0t} = \theta_t - \theta_0$, and

$$f(t) = \sum_{n=1}^{\infty} \sum_{k=-n}^0 F_{n,k} r_{0t}^k (-t)^{n+2k(\theta_t-\theta_0)},$$

$$g(t) = \sum_{n=1}^{\infty} \sum_{k=-n}^0 G_{n,k} r_{0t}^k (-t)^{n+2k(\theta_t-\theta_0)},$$

if $\Re(\theta_t - \theta_0) < \frac{1}{2}$.

On the **intersection point** of blue lines $\tilde{\mathcal{L}}_2^\infty$ and $\tilde{\mathcal{L}}_1^0$, we have $r_{0t} = 0$ and the generic asymptotics are **doubly truncated**,

$$\begin{aligned} f(t) &= \sum_{n=1}^{\infty} F_{n,0} (-t)^n, & F_{1,0} &= \frac{q^{\theta_0} - q^{-\theta_0}}{q^{\theta_t-\theta_0} - q^{\theta_0-\theta_t}}, \\ g(t) &= \sum_{n=1}^{\infty} G_{n,0} (-t)^n, & G_{1,0} &= \frac{q^{\theta_t} - q^{-\theta_t}}{q^{\theta_0-\theta_t} - q^{\theta_t-\theta_0}}. \end{aligned}$$

The latter power series solutions should be called Kaneko-Ohyama solutions ('13,'15).

black intersection points

Let (f, g) be the solution corresponding to the intersection point

$$\{\mathbf{x}_*(\mathbf{t})\} = \tilde{\mathcal{L}}_1^0 \cap \tilde{\mathcal{L}}_3^\infty,$$

and assume $\Re(\theta_0 - \theta_t), \Re(-\theta_0 - \theta_1) < \frac{1}{2}$, then $f(t)$ admits simultaneous uniformly convergent asymptotic expansions

$$f(t) = \frac{q^{\theta_0} - q^{-\theta_0}}{q^{\theta_0 - \theta_t} - q^{\theta_t - \theta_0}} t + tE_0(t) + \sum_{n=2}^{\infty} \sum_{k=0}^n f_{n,k} t^n E_0(t)^k \quad (t \rightarrow 0),$$

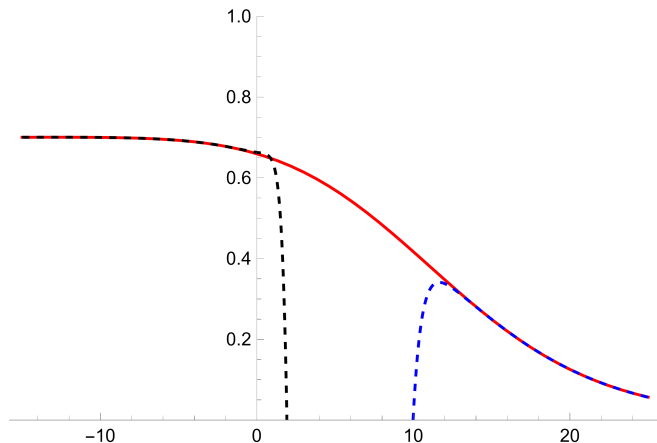
$$f(t) = \frac{q^{\theta_0} - q^{-\theta_0}}{q^{\theta_0 + \theta_1} - q^{-\theta_0 - \theta_1}} + E_\infty(t) + t^{-1} \sum_{n=2}^{\infty} \sum_{k=0}^n \dot{f}_{n,k} t^{-n} E_\infty(t)^k \quad (t \rightarrow \infty),$$

on compact sets $K \subseteq \mathbb{CP}^1 \setminus q^{\mathbb{Z} - 2\theta_0 + \theta_t - \theta_1}$, with $qK = K$, where

$$E_0(t) = c_0 \frac{\theta_q(q^{-\theta_t - \theta_1} t)}{\theta_q(q^{-2\theta_0 + \theta_t - \theta_1} t)}, \quad E_\infty(t) = c_\infty \frac{\theta_q(q^{-\theta_t - \theta_1} t^{-1})}{\theta_q(q^{+2\theta_0 - \theta_t + \theta_1} t^{-1})},$$

for some explicit constant factors c_0, c_∞ .

Plot of f on negative real line

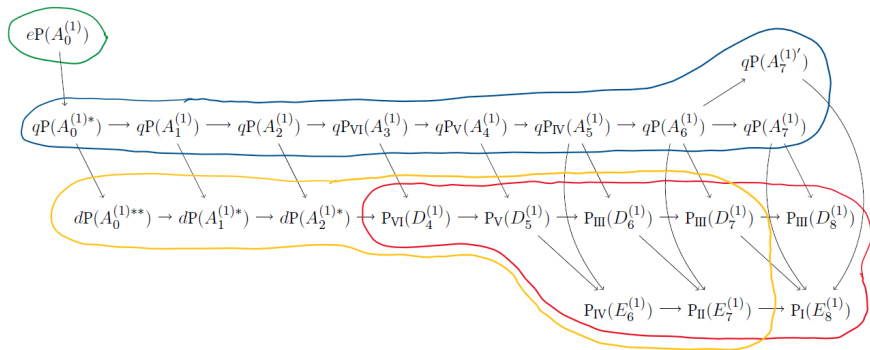


Plot of $f(-q^r)$ in red with $r \in (-15, 25)$ and parameter values

$$\theta_0 = \frac{1}{3}, \quad \theta_t = \frac{1}{5}, \quad \theta_1 = \frac{1}{7}, \quad \theta_\infty = \frac{1}{11}, \quad q = \exp\left(-\frac{1}{4}\right).$$

In dashed black and blue the series expansions around $t = -\infty$ and $t = 0$ respectively.

Sakai's classification



- In **green** the elliptic Painlevé equation.
- In **blue** the q -difference Painlevé equations.
- In **yellow** the additive Painlevé equations.
- In **red** the classical Painlevé equations.

Continuum limit 1/2

- The monodromy surface $\mathcal{M}_{q,\theta,t_0}$ is an embedded affine Segre surface of most **generic type**, in particular the divisor at infinity is an **elliptic curve** [Joshi, Mazzocco, PR 26’].
- Taking the limit $q \uparrow 1$, qP_{VI} reduces to P_{VI} .
- The limit of the Segre surface is also well-defined,

$$\lim_{q \uparrow 1} \mathcal{M}_{q,\theta,t_0} = \mathcal{M}_{1,\theta},$$

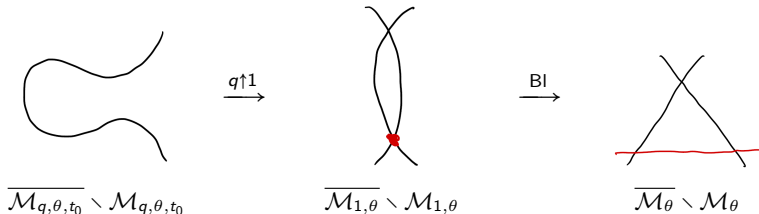
in the sense that the coefficients have well-defined limits. where the t_0 -dependence drops out in the limit.

- It is still an embedded affine Segre surface, but the divisor at infinity **degenerates** into **two conics** meeting at two distinct points.

Continuum limit 2/2

Theorem (Joshi, Mazzocco, PR 26')

*The limiting affine Segre surface $\mathcal{M}_{1,\theta}$ is isomorphic to the affine cubic surface $\mathcal{M}_\theta$ for P_{VI} . The isomorphism comes from a **blow-up map**, between their (canonical) projective completions, of one of the intersection points of the two conics at infinity of the Segre surface.*



Monodromy surfaces revisited

Del Pezzo (1887) studied algebraic surfaces with a degree d embedding in $\mathbb{CP}^d$. They can all be obtained by blowing up $9 - d$ points in $\mathbb{CP}^2$ in general position, $3 \leq d \leq 9$, except for the degree 8 case $\mathbb{CP}^1 \times \mathbb{CP}^1$.

In modern language **del Pezzo surfaces** are projective surfaces with ample anticanonical divisor class. (their degrees range from $1 \leq d \leq 9$.)

- Monodromy surface of $qP_{VI}(A_3^{(1)})$ is an affine del Pezzo surface of degree 4 with at infinity an elliptic curve. [Joshi, Mazzocco, PR '26].
- Monodromy surface of $qP_{IV}(A_5^{(1)})$ is an affine del Pezzo surface of degree 6 with at infinity an elliptic curve. [Joshi, PR '22/'23].
- Monodromy surface of P_{VI} is a del Pezzo surface of degree 3 with at infinity a triangle of lines.

For P_{IV} , P_{II} , P_I , alternative models for monodromy surfaces exist, each an embedded affine del Pezzo surface of a certain **degree** with a particular **cycle of curves** at **infinity**, consisting of only **smooth points** in its projective completion [PR and Stokes '26].

Sakai classification on the other side of RH?

Painlevé Eq.	Deg. of del Pezzo	At infinity
P_I	5	pentagon of lines
P_{II}	6	triangle of conics
P_{IV}	4	rectangle of lines
P_{VI}	3	triangle of lines
$qP_{IV}(A_5^{(1)})$	6	elliptic curve
$qP_{VI}(A_3^{(1)})$	4	elliptic curve

Conjecture (Stokes, PR '26)

Painlevé Eq.	Deg. of del Pezzo	At infinity
$qP(A_k^{(1)})$	$d = k + 1$	elliptic curve

Conclusion

- So far, we have done everything on a case by case basis. **General theory?**
- What is the **counterpart** of **Sakai's classification** on the other side of the Riemann-Hilbert correspondence?
- Does **Ramis' conjecture** on lines and truncated asymptotics also extend to all **discrete** Painlevé equations?
- What about beyond dimension 2? Embedded affine **Fano varieties** as monodromy varieties?